

The proof of Corollary 11.5 is incorrect. This was pointed out to me by Seval Özkurt. The following modification of Corollary 11.5 is correct and can still be used to prove Theorem III on the next page.

On page 72 replace the conclusion Corollary 11.5 by the inequality

$$|\phi(\alpha) - 1| \leq \frac{3}{(p-1)(|x|-1)}.$$

The proof of Corollary 11.5 remains the same except for the inequalities at the end. Replace these by

$$\begin{aligned} |\phi(\alpha) - 1| &= |\phi(\alpha_1) - \phi(\alpha_2)| \\ &\leq |\phi(\alpha_1) - \xi| + |\phi(\alpha_2) - \xi| \\ &\leq \frac{\|\theta_1\| + \|\theta_2\|}{q(|x|-1)} \\ &\leq \frac{3}{(p-1)(|x|-1)}. \end{aligned}$$

The proof of Theorem III on pages 73 and 74 should then be modified as follows. Replace the inequality in line +11 of page 73 by

$$|\phi(\alpha) - 1| \leq \frac{3}{(p-1)(|x|-1)}.$$

and replace the inequality in line +17 by

$$N(\alpha - 1) \leq \frac{2^{p-3}}{(p-1)^2} \frac{9}{(|x|-1)^2}.$$

Finally, replace the first ten lines of page 74 by the following:

“On the other hand, since $J(\alpha - 1)$ is an *integral* non-zero ideal, we have $N(J(\alpha - 1)) \geq 1$. Combining everything, we obtain the inequality

$$(|x| + 1)^{-\frac{p-1}{2q}\|\theta\|} \leq N(J)^{-1} \leq N(\alpha - 1) \leq \frac{2^{p-3}}{(p-1)^2} \frac{9}{(|x|-1)^2}.$$

Since $|x| \geq q^{p-1} > 20$, we have $|x| + 1 \leq \frac{4}{3}(|x| - 1)$ and hence

$$(|x| + 1)^{2 - \frac{p-1}{2q}\|\theta\|} \leq \frac{16}{(p-1)^2} 2^{p-3} \quad (*)$$

The size of θ satisfies $\|\theta\| \leq \frac{3q}{p-1}$, so that

$$(|x| + 1)^{1/2} \leq \frac{16}{(p-1)^2} 2^{p-3}.$$

By Corollary 6.5 we have $|x| > q^{p-1}$. Since $p \geq 5$, this leads to

$$q^{(p-1)/2} < (|x| + 1)^{1/2} \leq 2^{p-1},$$

contradicting $q \geq 5$.”